
KTU
NOTES

FAULT ANALYSIS

circuits).

conductor).

involve conductors-to-ground or short circuit
.

ntrolled by fuses or any device which does not
s, one or two phases of the circuit may be opened
es or phase is closed.

aracterised by increase in current and fall in
ncy .

aracterised by increase in voltage and frequency

the faulted phases.

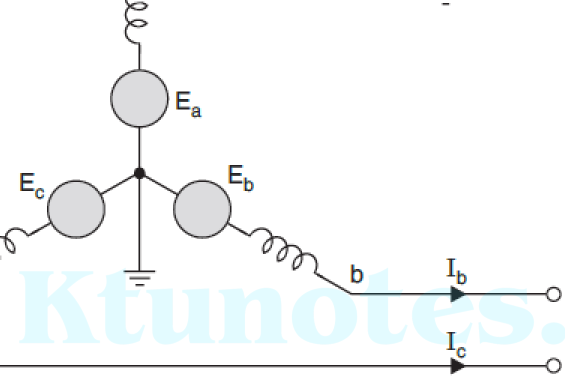
t;

und fault;

ree are the unsymmetrical faults as the symmetry is
two

netrical components will be utilized to analyse the
ystem.

a balanced fault which could also be analysed using
nents.



y grounded, unloaded alternator: L-G fault on phase a.

ace on phase a.

ns are $V_a = 0$

$I_b = 0$

$I_c = 0$

$$I_f = I_a$$

$$= \frac{1}{3}(I_a + \lambda I_b + \lambda^2 I_c)$$

$$= \frac{1}{3}(I_a + \lambda^2 I_b + \lambda I_c)$$

$$= \frac{1}{3}(I_a + I_b + I_c)$$

$$= I_{a_2} = I_{a_0} = I_a/3$$

work equations are

$$V_{a_0} = - I_{a_0} Z_0$$

$$V_{a_1} = E_a - I_{a_1} Z_1$$

$$V_{a_2} = - I_{a_2} Z_2$$

es of V_{a_0}, V_{a_1} and V_{a_2} from the sequence network equation,

$$a - I_{a_1} Z_1 - I_{a_2} Z_2 - I_{a_0} Z_0 = 0$$

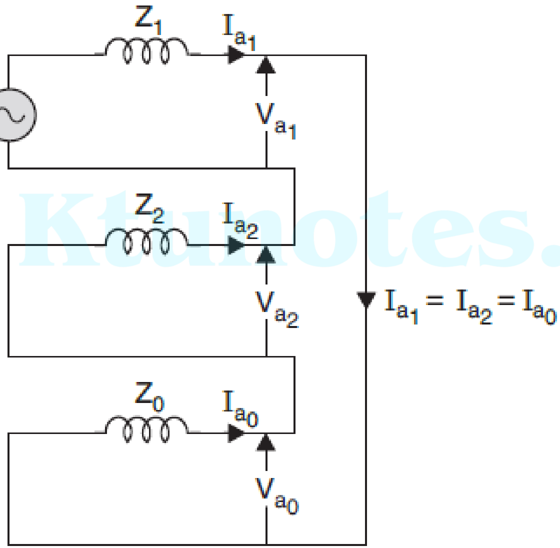
$$I_{a_1} = I_{a_2} = I_{a_0}$$

$$a - I_{a_1} Z_1 - I_{a_1} Z_2 - I_{a_1} Z_0 = 0$$

$$I_{a_1} = \frac{E_a}{Z_1 + Z_2 + Z_0}$$

$$I_a = 3I_{a_1}$$

sequence networks must be connected in series.



Interconnection of sequence networks for L-G fault.

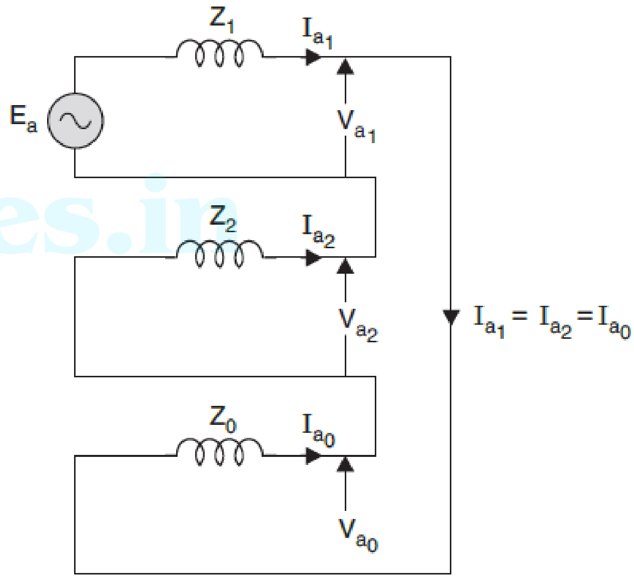
rrent and the line-to-line voltages. Neglect resistance.

ally the positive sequence
han the negative sequence
sitive sequence impedance
transient state, it may be
sequence impedance. The
r a line-to-ground fault
g. E.13.3.

eutral voltage at the fault
e 1.0 + j0.0 p.u. For a line-
lt impedance is

$$.25 + j0.35 + j0.1 = j0.7$$

$$\frac{\quad_a}{\quad_2 + Z_0} = \frac{1 + j0.0}{j0.7} = -j1.428$$



Interconnection of sequence network.

the base current = $\frac{25 \times 1000}{\sqrt{3} \times 13.2}$ = 1093 amps

current in amperes = 1093 × 4.285 = 4685 amps

voltages, we first find out the sequence components of voltages.

$$V_{a_1} = E_a - I_{a_1} Z_1$$

$$= 1 + j0.0 - (-j1.428)(j0.25)$$

$$= 1 - 0.357 = 0.643$$

$$V_{a_2} = - I_{a_2} Z_2 = -(-j1.428)(j0.35)$$

$$= - 0.4998$$

$$V_{a_0} = - I_{a_0} Z_0 = -(-j1.428)(j0.1) = 0.1428$$

heck $V_a = 0$. Substituting the values of V_{a_1}, V_{a_2} and V_{a_0},

$$0.643 - 0.4998 - 0.1428 \simeq 0$$

$$V_b = V_{b_1} + V_{b_2} + V_{b_0} \text{ and } V_c = V_{c_1} + V_{c_2} + V_{c_0}$$

$$V_{b_1} = \lambda^2 V_{a_1} = (-0.5 - j0.866)(0.643)$$

$$= - 0.3215 - j0.5568$$

$$V_{b_2} = \lambda V_{a_2} = (-0.5 + j0.866)(-0.50)$$

$$= (0.25 - j0.433)$$

$$V_{c_1} = \lambda V_{a_1} = (-0.5 + j0.866)(0.643)$$

$$= -0.3215 + j0.5568$$

$$V_{c_2} = \lambda^2 V_{a_2} = (-0.5 - j0.866)(-0.5)$$

$$= 0.25 + j0.433$$

$$V_b = -0.3215 - j0.5568 + 0.25 - j0.433 - 0.1428$$

$$= -0.2143 - j0.9898$$

$$V_c = -0.3215 + j0.5568 + 0.25 + j0.433 - 0.1428$$

$$= -0.2143 + j0.9898$$

ine-to-line voltage

$$V_{ab} = V_a - V_b. \text{ Since } V_a = 0,$$

$$V_{ab} = -V_b = 0.2143 + j0.9898$$

$$V_{ac} = -V_c = 0.2143 - j0.9898$$

$$V_{bc} = V_b - V_c = -j2 \times 0.9898$$

$$= -j1.9796$$

$$V_{ab} = 0.2143 + j0.9898 = \sqrt{(0.4592 + 9.797) \times 10^{-1}}$$

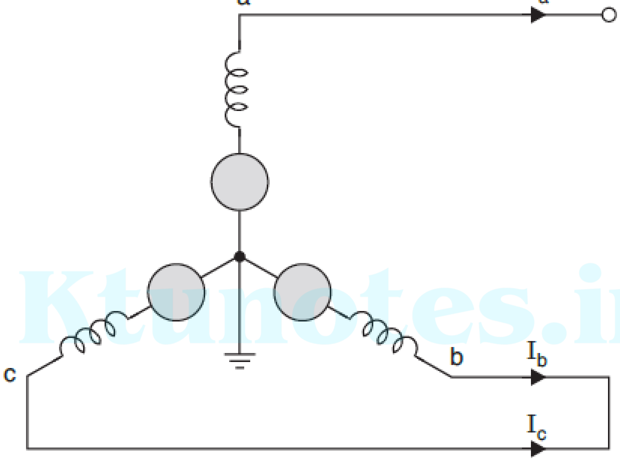
$$= \sqrt{10.346 \times 10^{-1}} = \sqrt{1.0346} = 1.0127 \text{ p.u.}$$

to-line voltage will be

$$V_{ab} = 1.0127 \times \frac{13.2}{\sqrt{3}} = 7.717 \text{ kV}$$

$$V_{ac} = 7.717 \text{ kV}$$

$$V_{bc} = 1.9796 \times \frac{13.2}{\sqrt{3}} = 15.08 \text{ kV.}$$



: L-L fault on an unloaded and neutral grounded alternator.

lt takes place on phases b and c

ditions are

$$I_a = 0$$
$$I_b + I_c = 0$$
$$V_b = V_c$$

$$I_b = -I_c$$

$$I_{a_2} = \frac{1}{3}(I_a + \lambda^2 I_b + \lambda I_c)$$

$$I_{a_0} = \frac{1}{3}(I_a + I_b + I_c)$$

g for I_a, I_b and I_c

$$I_{a_1} = \frac{1}{3}(0 + \lambda I_b - \lambda^2 I_b)$$

$$= \frac{1}{3}(\lambda - \lambda^2) I_b$$

$$I_{a_2} = \frac{1}{3}(0 + \lambda^2 I_b - \lambda I_b)$$

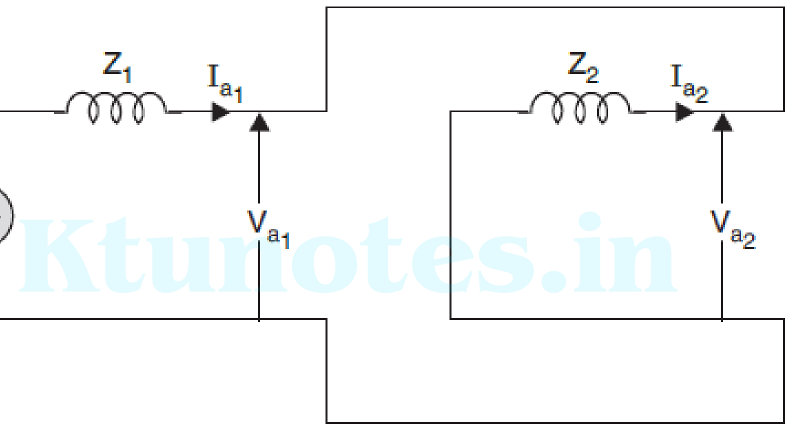
$$= \frac{I_b}{3}(\lambda^2 - \lambda)$$

$$I_{a_0} = \frac{1}{3}(0 + 0) = 0$$

mponent of current is equal in magnitude but
negative sequence component of current.

$$I_{a_1} = -I_{a_2}$$

condition zero sequence network is not required
negative-sequence networks are to be connected



11 Interconnection of sequence networks for L-L fault.

rrent $I_f = I_b = I_{b0} + I_{b1} + I_{b2}$

$$I_{b0} = 0 \quad \because \quad I_{a0} = 0$$

Also $I_{b1} = a^2 I_{a1}$

$$I_{b2} = a I_{a2}$$

$$\because \quad I_{a2} = -I_{a1}, \; I_{b2} = -a I_{a1}$$

$$\therefore \quad I_f = I_b = a^2 I_{a1} - a I_{a1}$$

$$= (a^2 - a) I_{a1}$$

$$= (-0.5 - j0.866 + 0.5 - j0.866) I_a$$

$$= -j1.732 I_{a1}$$

$$I_f = -j\sqrt{3} \times I_{a1}$$

$$V_c = V_{a_0} + \lambda V_{a_1} + \lambda^2 V_{a_1}$$

relatio

$$+ \lambda V_{a_2} = V_{a_0} + \lambda V_{a_1} + \lambda^2 V_{a_2}$$

$$- \lambda) V_{a_1} = (\lambda^2 - \lambda) V_{a_2}$$

$$V_{a_1} = V_{a_2}$$

quence component of voltage equals the negative-sequence component ns that the two sequence networks are connected in opposition. Now ce network equation and the equation (13.32),

$$V_{a_1} = V_{a_2}$$

$$- I_{a_1} Z_1 = - I_{a_2} Z_2 = I_{a_1} Z_2$$

$$I_{a_1} = \frac{E_a}{Z_1 + Z_2}$$

t current and the line-to-line voltage at the fault
fault occurs at the terminals of the alternator
ous Example

equence network for L-L
E.13.4. Since the zero
sent, assuming $(1 + j0.0)$
ge,

$$\frac{E_a}{{}_1 + Z_2} = \frac{1 + j0.0}{j0.25 + j0.35}$$

$$\frac{\cdot\, j0.0}{j0.6} = -j1.667$$

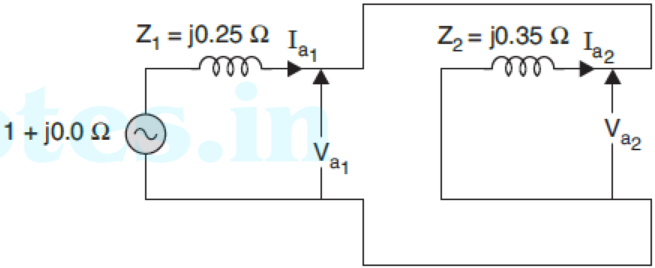


Fig. E.13.4 Sequence network.

ault current, $I_b = -I_c$, we use the following relations:

$$I_b = I_{b_1} + I_{b_2} + I_{b_0} = I_{b_1} + I_{b_2} = \lambda^2 I_{a_1} + \lambda I_{a_2}$$

$$= (-0.5 - j0.866)(-j1.667) + (-0.5 + j0.866)(j1.667)$$

$$= j0.833 - 1.4436 - j0.833 - 1.4436$$

$$= -2.8872 \text{ p.u.}$$

nt is 1093 amperes.

current = 1093 × 2.8872 = 3155.71 amperes

to-line voltage we find out the sequence components of voltages

$$V_{a_1} = E_a - I_{a_1} Z_1 = 1 + j0.0 - (-j1.667)(j0.25)$$

$$= 1 - 0.4167 = 0.5833$$

$$V_{a_2} = -I_{a_2} Z_2 = (-j1.667)(j0.35) = 0.5834$$

$$V_{a_1} = V_{a_2} = 0.5833 \text{ p.u.}$$

$$V_a = V_{a_1} + V_{a_2} + V_{a_0} = V_{a_1} + V_{a_2} = 2 \times 0.5833 = 1.1666 \text{ p.u.}$$

$$V_b = \lambda^2 V_{a_1} + \lambda V_{a_2}$$

$$= (-0.5 - j0.866)(0.5833) + (-0.5 + j0.866)(0.5833)$$

$$= -0.5833$$

$$= -0.5833$$

$$V_b = V_c = -0.5833$$

tage

$$V_{ab} = V_a - V_b = 1.1666 - (-0.5833) = 1.7499$$

$$V_{ac} = V_a - V_c = 1.7499$$

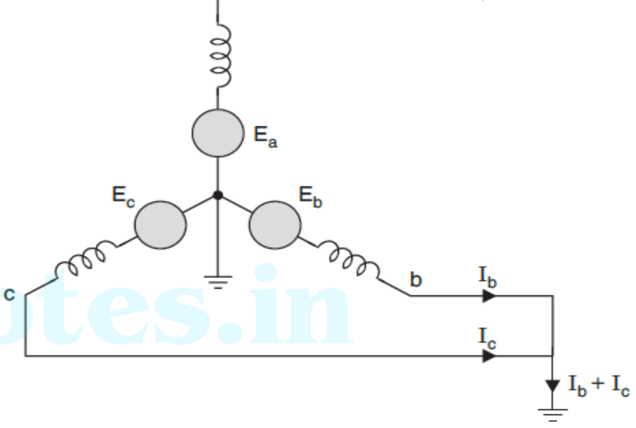
$$V_{bc} = V_b - V_c = 0.0$$

-to-line voltage

$$V_{ab} = 1.7499 \times \frac{13.2}{\sqrt{3}} = 13.33 \text{ kV}$$

$$V_{ac} = 13.33 \text{ kV}$$

$$V_{bc} = 0.0 \text{ kV.} \quad \textbf{Ans.}$$



.. solidly grounded, unloaded alternator, L-L-G fault.

und fault takes place on phases b and c

ditions are

$$I_a = 0$$
$$V_b = 0$$
$$V_c = 0$$

$$= V_a/3$$

$$V_{a_2} = \tfrac{1}{3}(V_a + \lambda^2 V_b + \lambda V_c)$$

$$= V_a/3$$

$$V_{a_0} = V_{a_1} = V_{a_2}$$

ages and substituting in the sequence network equations

$$V_{a_0} = V_{a_1}$$

$$- I_{a_0} Z_0 = E_a - I_{a_1} Z_1$$

$$I_{a_0} = -\frac{E_a - I_{a_1} Z_1}{Z_0}$$

$$V_{a_2} = V_{a_1}$$

$$- I_{a_2} Z_2 = E_a - I_{a_1} Z_1$$

$$I_{a_2} = -\frac{E_a - I_{a_1} Z_1}{Z_2} \qquad (1$$

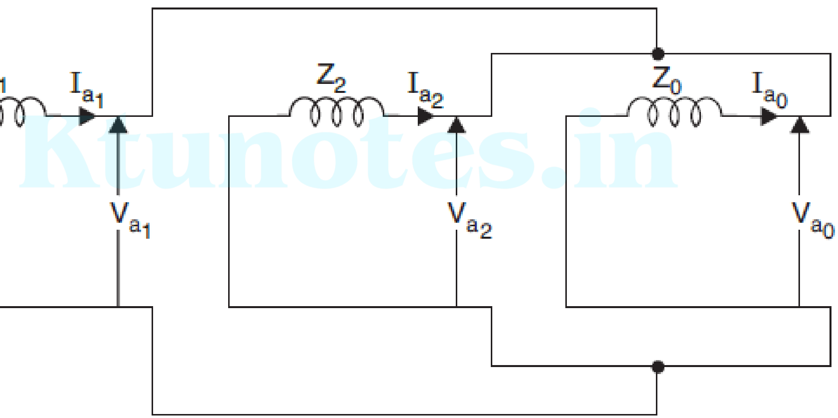
$$I_a = I_{a_1} + I_{a_2} + I_{a_0} = 0$$

g values of I_{a_2} and I_{a_0} from equations

$$I_{a_1} - \frac{E_a - I_{a_1} Z_1}{Z_2} - \frac{E_a - I_{a_1} Z_2}{Z_0} = 0$$

g the terms gives

$$I_{a_1} = \frac{E_a}{Z_1 + \frac{Z_0 Z_2}{Z_0 + Z_2}}$$



Interconnection of sequence networks for L-L-G fault.

$$= 2I_{a0} + I_{a1}\left[a^2 + a\right] + I_{a2}\left[a^2 + a\right]$$

he relation

$$1 + a + a^2 = 0,$$

$$a + a^2 = -1$$

$= -1$ in the above eqn. we get,

$$I_f = 2I_{a0} - I_{a1} - I_{a2}$$

$$= 2I_{a0} - (I_{a1} + I_{a2})$$

1)

$$I_a = I_{a0} + I_{a1} + I_{a2} = 0$$

$$I_{a1} + I_{a2} = -1_{a0}$$

$= V_{a2} = V_{a0}$ by equation (4), all the sequence networks are to be connected in s shown in figure.

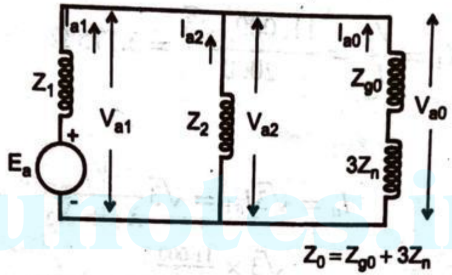


Fig. 3.22 Connection Diagram.

he figure, we can also verify that

$$I_{a1} + I_{a2} + I_{a0} = I_a = 0.$$

connection diagram,we can write

$$I_{a0} = -I_{a1} \times \frac{Z_2}{Z_2 + Z_0}$$

fault current in LLG fault is given by

$$I_f = 3 \times I_{a0}$$

t current and the line-to-line voltages at the fault

to-ground fault occurs at the terminals of the
in previous Example

uming $(1 + j0.0)$ p.u. as prefault voltage,

$$I_{a_1} = \frac{E_a}{Z_1 + \frac{Z_0\,Z_2}{Z_0 + Z_2}} = \frac{1 + j0.0}{j0.25 + \frac{j0.1 \times j0.35}{j0.45}} = \frac{1 + j0.0}{j0.25 + j0.0778}$$

$$= \frac{1 + j0.0}{j0.3278} = -j3.0506 \text{ p.u.}$$

, $V_{a_1} = V_{a_2} = V_{a_0}$

$$V_{a_1} = E_a - I_{a_1} Z_1$$

and I_{a_0}, we should first find V_{a_1} and since $V_{a_1} = V_{a_2} = -I_{a_2} Z_2$, I_{a_2} can be

$_1 = V_{a_0} = -I_{a_0} Z_0$, I_{a_0} can be obtained.

$$I_{a_2} = -\frac{V_{a_2}}{Z_2} = -\frac{0.2374}{j0.35} = j\frac{0.2374}{0.35} = j0.678$$

$$I_{a_0} = -\frac{V_{a_0}}{Z_0} = -\frac{0.2374}{j0.1} = j2.374$$

$$I_{a_2} + I_{a_0} = j0.678 + j2.374 = j3.05 = -I_{a_1}$$

fault current $= I_b + I_c = 3I_{a_0} = 3 \times j2.374 = j7.122$ p.u.

current is 1093 amperes, the fault current will be

$$1093 \times 7.122 = 7784.3 \text{ amperes}$$

$$V_a = V_{a_1} + V_{a_2} + V_{a_0} = 3V_{a_1} = 3 \times 0.2374 = 0.7122$$

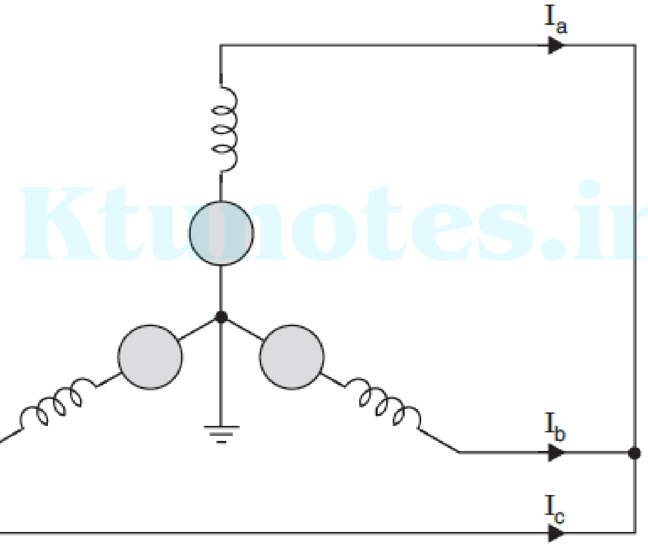
$$V_b = V_c = 0$$

-line fault voltage,

$$V_{ab} = V_a = 0.7122 \times \frac{13.2}{\sqrt{3}} = 5.42 \text{ kV}$$

$$V_{ac} = V_a = 0.7122 \times \frac{13.2}{\sqrt{3}} = 5.42 \text{ kV}$$

$$V_{bc} = 0.0 \text{ kV}$$



: A 3-phase neutral grounded and unloaded alternator 3-phase shorted.

$$V_a = V_b = V_c$$

g the relation

$$I_{a_1} = \tfrac{1}{3}(I_a + \lambda I_b + \lambda^2 I_c)$$

uting the values of I_b and I_c,

$$I_{a_1} = \tfrac{1}{3}(I_a + \lambda^3 I_a + \lambda^3 I_a)$$

$$= I_a$$

$$I_{a_2} = \tfrac{1}{3}(I_a + \lambda^2 I_b + \lambda I_c)$$

tituting for I_b and I_c in terms of I_a,

$$I_{a_2} = \tfrac{1}{3}(I_a + \lambda^4 I_a + \lambda^2 I_a)$$

$$= \tfrac{1}{3}(I_a + \lambda I_a + \lambda^2 I_a)$$

$$= \frac{I_a}{3}(1 + \lambda + \lambda^2)$$

$$= 0$$

larly, $$I_{a_0} = \tfrac{1}{3}(I_a + I_b + I_c)$$

$$= 0$$

nce component of current is equal to the phase e voltage boundary relation,

$$V_{a_1} = \tfrac{1}{3}(V_a + \lambda V_b + \lambda^2 V_c) = \tfrac{1}{3}(V_a + \lambda V_a + \lambda^2 V_a)$$

$$= \frac{V_a}{3}(1 + \lambda + \lambda^2) = 0$$

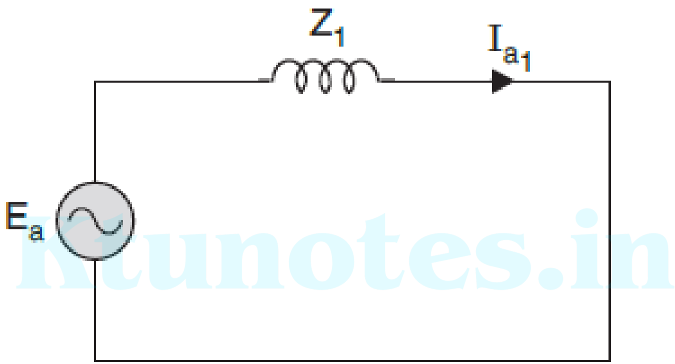
$$V_{a_2} = \tfrac{1}{3}(V_a + \lambda^2 V_b + \lambda V_c)$$

$$= 0$$

$$V_{a_0} = 0$$

$$V_{a_1} = 0 = E_a - I_{a_1} Z_1,$$

$$I_{a_1} = \frac{E_a}{Z_1}$$



Interconnection of sequence network-3-phase fault.

currents are present in all types of faults.

currents are present in all unsymmetrical faults.

ents are present when the neutral of the system is
lt also involves the ground.

eutral current is equal to $3I_{a_0}$

$$A = \frac{Base\ MVA}{Equivalent\ PU\ impedance}$$

$$\frac{\text{ual value of the quantity (in any unit)}}{\text{se or reference value in the same unit}}$$

ring the three basic quantities are voltage, current

nce

$$\text{se} = \frac{V_{base}}{VA_{base}/V_{base}} = \frac{{V_{base}}^2}{VA_{base}}$$

$$\text{actual} \times \frac{MVA_{base}}{{kV_{base}}^2}$$

$$_{\text{new}} = Z_{\text{p.u. old}} \cdot \frac{kVA_{\text{new}}}{kVA_{\text{old}}} \cdot \left(\frac{V_{\text{old}}}{V_{\text{new}}}\right)^2$$

pu value*100

of the transformer referred to primary side be Z_p and that on the
n

$$Z_p = Z_s \left(\frac{V_p}{V_s}\right)^2$$

primary and secondary voltages of the transformer.

$$Z_{p\text{ p.u.}} = \frac{Z_p I_p}{V_p} = Z_s \left(\frac{V_p}{V_s}\right)^2 . \frac{I_p}{V_p}$$

$$= Z_s . \frac{V_p I_p}{V_s^2} = Z_s . \frac{V_s I_s}{V_s^2} = \frac{Z_s I_s}{V_s}$$

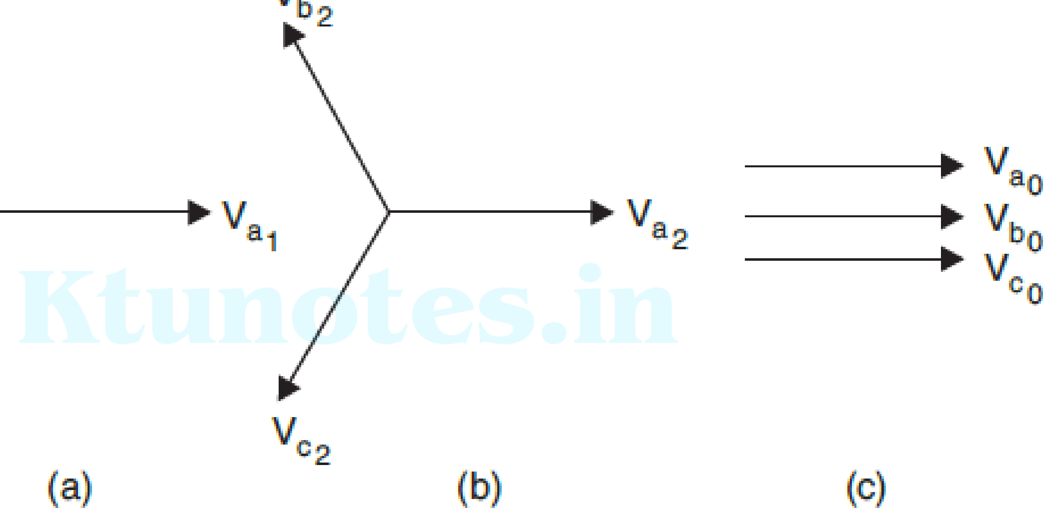
$$= Z_{s\text{ p.u.}}$$

s of vectors.

ce component which has three vectors of
e but displaced in phase from each other by
e same phase sequence as the original

nce component which has three vectors of
e but displaced n phase from each other by
e phase sequence opposite to the original

omponent which has three vectors of equal
also are in phase with each other.



(*a*) Positive sequence component; (*b*) Negative sequence component; (*c*) Zero sequence component.

shown in figure 4.1.1.

The positive, negative and zero sequence reactances of each being respectively, j0.18, j0.15, j0.10 p.u. The star point of one of the generator is isolated and that of the other is earthed through a 2.0 Ω resistor. A single line-to-ground fault occurs at the t of the generators. Estimate (i) fault current, (ii) current in grounded resistor across grounding resistor.

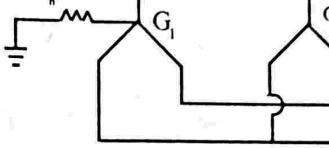


Fig 4.1.1

SOLUTION

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 20\,MVA \quad ; \quad kV_b = 11\,kV$$

$$\text{Base impedance, } Z_b = \frac{kV_b^2}{MVA_b} = \frac{11^2}{20} = 6.05\ \Omega$$

$$\text{p.u. value of neutral resistance} = \frac{\text{Actual value}}{\text{Base impedance}} = \frac{2}{6.05} = 0.3306\ \text{p.u.}$$

The single line diagram and the sequence networks of the given p shown in fig (4.1.2) to (4.1.5)

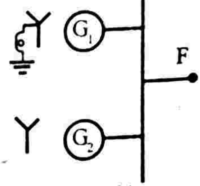


Fig 4.1.2 : Single line diagram

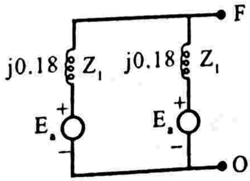


Fig 4.1.3 : Positive sequence network

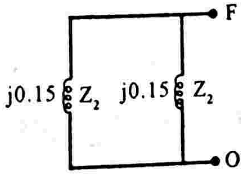
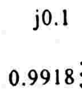


Fig 4.1.4 : Negative sequence network



F se

Hence the thevenin's equivalent of sequence networks are connected in series as shown in fig (4.1.9). The fault current is calculated by taking the prefault voltage $V_{pf} = 1$ p.u.

From fig 4.1.9. we get,

$$I_{a1} = \frac{1\angle 0°}{j0.09 + j0.075 + 0.9918 + j0.1}$$

$$= \frac{1}{0.9918 + j0.265} = \frac{1}{1.0266\angle 15°} = 0.9741\angle -15° \text{ p.u.}$$

i. To find fault current

Fault current, $I_f = I_a = 3\, I_{a1}$

V_{pf} =

$$= 3 \times 0.9741\angle -15°$$

$$= 2.9223\angle -15° \text{ p.u.}$$

Base current, $I_b = \frac{kVA_b}{\sqrt{3}\, kV_b} = \frac{20 \times 1000}{\sqrt{3} \times 11} = 1049.7$ A

$\therefore$ Actual value of fault current = p.u value of fault current × Base curre

$$= 2.9223\angle -15° \times 1049.7 = 3067.5\angle -15°$$ A

ii. To find the current through neutral resistor

The current through the neutral resistor is same as that of fault c

$\therefore$ Current through neutral resistor = $2.9223\angle -15°$ p.u. or 3.06

iii. To find the voltage across grounding resistor

From the thevenin's equivalent of zero sequence network we ge

The voltage across grounding resistor $= 3\, R_n\, I_{a0} = 3\, R_n\, I_{a1}$

$$= 3 \times 0.3306 \times 0.9741\angle -15° = 0.$$

A salient - pole generator without dampers is rated 20 MVA, axis subtransient reactance of 0.25 per unit. The negative and zero sec and 0.10 per unit respectively. The neutral of the generator is solidly sub-transient current in the generator and the line-to-line voltages fo when a single line-to-ground fault occurs at the generator terminals unloaded at rated voltage. Neglect resistance.

SOLUTION

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 20\,MVA \quad ; \quad kV_b = 13.8\,kV$$

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{20 \times 1000}{\sqrt{3} \times 13.8} = 836.7\ A$$

Given that, $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and Z_0

For a single line-to-ground fault in an unloaded generator t connected in series as shown in fig 4.2.1. The following relations exi single line-to-ground fault.

$$I_{a0} = I_{a1} = I_{a2} \quad ; \quad I_f = I_a = 3I_{a1}$$

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0}$$

Here E_a' is phase value of induced emf. Let it be (i.e. $E_a = 1\angle 0°$ p.u.)

$$\therefore I_{a1} = \frac{1}{j0.25 + j0.35 + j0.1} = \frac{1}{j0.7} = -j1.43 = 1.43\angle -90°\ \text{p.u}$$

Subtransient fault current in p.u., $I_f'' = |I_a| = 3\,|I_{a1}| = 3 \times$
$= 4.29$ p.u.

Actual value of subtransient fault current = p.u. value of fault × Ba
$= 4.29 \times 836.7 = 3589.4$ A

$$\begin{bmatrix} V_c \end{bmatrix} \begin{bmatrix} 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a2} \end{bmatrix}$$

where $a = 1\angle 120° = -0.5 + j0.866$
and $a^2 = 1\angle 240° = -0.5 - j0.866$

$$V_a = V_{a0} + V_{a1} + V_{a2} = 0.643 - 0.50 - 0.143 = 0$$

$$V_b = V_{a0} + a^2 V_{a1} + aV_{a2}$$

$$= -0.143 + 0.643\,(-0.5 - j0.866) - 0.50(-0.5 + j0.866)$$

$$= -0.143 - 0.322 - j0.557 + 0.25 - j0.433 = -0.215 - j0.990$$

$$V_c = V_{a0} + aV_{a1} + a^2 V_{a2}$$

$$= -0.143 + 0.643(-0.5 + j0.866) - 0.5(-0.5 - j0.866)$$

$$= -0.143 - 0.322 + j0.557 + 0.25 + j0.433 = -0.215 + j0.99 \text{ p}$$

Line-to-line voltages are

$$V_{ab} = V_a - V_b = 0.215 + j0.990 = 1.01\angle 77.7° \text{ p.u.}$$

$$V_{bc} = V_b - V_c = -0.215 - j0.99 + 0.215 - j0.99 = -j1.980 = 1.98$$

$$V_{ca} = V_c - V_a = -0.215 + j0.99 = 1.01\angle 102.3° \text{ p.u.}$$

Since the generated voltage to neutral E_a was taken as 1.0 p.u voltages are expressed in p.u. on the base of phase voltage. The postfaul in volts are obtained by multiplying the p.u. value with base value of

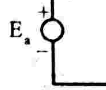
$$\text{Line value of base voltage} = 13.8 \text{ kV}$$

$$\text{Phase value of base voltage} = \frac{13.8}{\sqrt{3}} \text{ kV} = 7.97 \text{ kV}$$

$$V_{ab} = 1.01 \times 7.97\angle 77.7° = 8.05\angle 77.7° \text{ kV}$$

$$V_{bc} = 1.98 \times 7.97\angle 270° = 15.78\angle 270° \text{ kV}$$

$$V_{ca} = 1.01 \times 7.97\angle 102.3° = 8.05\angle 102.3° \text{ kV}$$



$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{20 \times 1000}{\sqrt{3} \times 13.8}$$

$$= 836.7 \text{ A} \approx 837 \text{ A}$$

Given that Z_1 = j0.25 p.u. . ; Z_2 = j0.35 p.u. and Z_0 = j0.1 p.u

For a line-to-line fault the positive and negative sequence netw

parallel as shown in fig 4.3.1.

The following relations exists for a single line-to-line fault.

$$I_a = 0 \quad ; \quad I_b = -I_c \quad ; \quad I_{a0} = 0 \quad ; \quad I_{a2} = -I_{a1} \quad ; \quad V$$

From fig 4.3. we get,

$$I_{a1} = \frac{E_a}{Z_1 + Z_2}$$

Here E_a is phase value of induced emf. Let it be 1 p.u. (i.e. E_a =

$$\therefore I_{a1} = \frac{1}{j0.25 + j0.35} = \frac{1}{j0.6} = -j1.667 = 1.667\angle -90° \text{ p.u.}$$

$$I_{a2} = -I_{a1} = j1.667 \text{ p.u.}$$

$$I_{a0} = 0$$

The line currents are related to symmetrical components by the

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

where $a = 1\angle 120° = -0.5 + j0.866$

and $a^2 = 1\angle 240° = -0.5 - j0.866$

$$I_a = I_{a0} + I_{a1} + I_{a2} = -j\,1.667 + j1.667 = 0$$

From the sequence networks of the generator the symmet
phase-a voltage are calculated as shown below.

$$V_{a0} = 0 \quad ; \quad V_{a1} = E_a - I_{a1} Z_1 \quad ; \quad V_{a2}$$

$$\therefore V_{a1} = V_{a2} = 1-(-j1.667)(j0.25) = 1-0.417 = 0.583 \text{ p.}$$

The phase voltages are given by,

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$V_a = V_{ao} + V_{a1} + V_{a2} = 0.583 + 0.583 = 1.166\angle 0° \text{ p.u.}$$

$$V_b = V_{a0} + a^2 V_{a1} + aV_{a2}$$

$$V_b = 0.583(-0.5 - j0.866) + 0.583(-0.5 + j0.866) = -0.583 \text{ p.u.}$$

$$V_c = V_b = -0.583 \text{ p.u.}$$

Line voltages are

$$V_{ab} = V_a - V_b = 1.166 + 0.583 = 1.749\angle 0° \text{ p.u.}$$

$$V_{bc} = V_b - V_c = -0.583 + 0.583 = 0 \text{ p.u.}$$

$$V_{ca} = V_c - V_a = -0.583 - 1.166 = 1.749\angle 180° \text{ p.u.}$$

Since the generated phase voltage E_a is taken as 1 p.u., the a
expressed in p.u. on the base of phase voltage. The post fault line vol
are obtained by multiplying the p.u. value with the base value of phas

Line value of base voltage = 13.8 kV

$$\text{Phase value of base voltage} = \frac{13.8}{\sqrt{3}} = 7.9 \text{ kV}$$

$$V_{ab} = 1.749\angle 0° \times 7.97 = 13.94\angle 0° \text{ kV} \quad ; \quad V$$

$$V_{ca} = 1.749\angle 180° \times 7.97 = 13.94\angle 180° \text{ kV}$$

Base current, $I_b = \dfrac{kVA_b}{\sqrt{3}\, kV_b} = \dfrac{20\times 10^3}{\sqrt{3}\times 13.8} = 836.7A \approx 837A$

Given that $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a double line-to-ground fault the positive, negative and zero sequence connected in parallel as shown in fig 4.4.1.

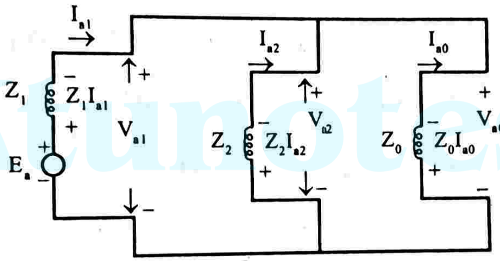


Fig 4.4.1

The following relations exists for a double line-to-ground fault.

$$I_{a1} = -(I_{a2} + I_{a0}) \quad ; \quad V_{a0} = V_{a1} = V_{a2} \qquad ; \quad V_b = V_c = 0$$

From fig 4.4.1 we can calculate the symmetrical components of phas current as shown below.

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 Z_0 / (Z_2 + Z_0)} = \frac{1.0 + j0}{j0.25 + (j0.35 \times j0.10) / (j0.35 + j0.1}$$

$$= \frac{1.0}{j0.25 + j0.0778} = \frac{1.0}{j0.3278} = -j3.05 \text{ p.u.}$$

$$V_{a1} = V_{a0} = V_{a2} = E_a - I_{a1} Z_1 = 1 - (-j3.05)\,(j0.25) = 1.0 - 0.763 =$$

$$I_{a2} = \frac{-V_{a2}}{Z_2} = -\frac{0.237}{j0.35} = j0.68 \text{ p.u}$$

$$I_{a0} = \frac{-V_{a0}}{Z_0} = \frac{-0.237}{j0.10} = j2.37 \text{ p.u}$$

$$I_c = I_{a0} + aI_{a1} + a^2 I_{a2}$$

$$= j2.37 + (-0.5 + j0.866)(-j3.05) + (-0.5 - j0.866)(j0.68$$

$$= j1.525 + 2.641 - j0.34 + 0.589 + j2.37 = 3.230 + j3.555$$

$$= 4.80\angle 47.7^\circ \text{ p.u.}$$

Fault current, $I_f = I_b + I_c = -3.230 + j3.555 + 3.230 + j3.555 = j7.$

The phase and line voltages are calculated as shown below

$$V_a = V_{a1} + V_{a2} + V_{a0} = 3V_{a1} = 3 \times 0.237 = 0.711 \text{ p.u.}$$

$$V_b = V_c = 0$$

$$V_{ab} = V_a - V_b = 0.711 \text{ p.u.}$$

$$V_{bc} = 0$$

$$V_{ca} = V_c - V_a = -0.711 \text{ p.u.}$$

The actual values of line currents and voltages are calculated a

$$I_a = 0$$

$$I_b = 837 \times 4.8\angle 132.3^\circ = 4017\angle 132.3^\circ \text{ A}$$

$$I_c = 837 \times 4.8\angle 47.7^\circ = 4017\angle 47.7^\circ \text{ A}$$

$$I_f = 837 \times 7.11\angle 90^\circ = 5951\angle 90^\circ \text{ A}$$

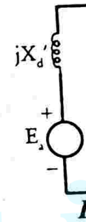
$$V_{ab} = 0.711 \times 7.97 = 5.66\angle 0^\circ \text{ kV}$$

$$V_{bc} = 0$$

$$V_{ca} = -0.711 \times 7.97 = 5.66\angle 180^\circ \text{ kV}$$

Case(i) : 3-phase fault

The circuit model used to calculate the transient fault current for a 3-pha
generator is shown in fig 4.5.1



Given that, $E_g' = 1$ p.u. , $I' = 3.33$ p.u.

When load current is neglected, $E_g'' = E_g' = E_g = E_a$

$$\therefore X_d' = \frac{|E_a|}{|I'|} = \frac{1}{3.33} = 0.3 \text{ p.u.}$$

We know that the reactance during symmetrical fault is positive sequence

$$\therefore \text{positive sequence reactance of generator, } X_1 = X_d' = 0.3 \text{ p.u.}$$

Case (ii) : L - L fault

For a line-to-line fault on a generator the positive and negative sequence networks can be connected in parallel as shown in fig 4.5.2

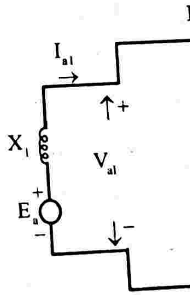


Fig :

From fig 4.5.2 we get

$$I_{a1} = \frac{E_a}{jX_1 + jX_2} \quad .....(4.5.1)$$

$$\therefore |I_{a1}| = \frac{E_a}{X_1 + X_2}$$

Let us assume that the line-to-line fault is due to a short between phas
Hence the fault current is I_b.

From the symmetrical components of the current we can write

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

1.732

From equation (4.5.1) & (4.5.2) we can write,

$$1.2875 = \frac{E_a}{X_1 + X_2}$$

$$\therefore X_1 + X_2 = \frac{E_a}{1.2875} \quad ; \quad \therefore X_2 = \frac{E_a}{1.2875} - X_1 = \frac{1}{1.2875} - 0.3 =$$

Case (iii) L-G fault

For a single line-to-ground fault, on a generator the sequence netw
series as shown in fig 4.5.3. For a single line-to-ground fault, the fault c
equal to $3|I_{a1}|$.

Given that, $I_f = 3.01$ p.u.

$$\therefore 3\,|I_{a1}| = I_f = 3.01 \quad \text{or} \quad |I_{a1}| = \frac{3.01}{3} = 1.0033 \text{ p.u.}$$

From fig 4.5.3 we can write

$$I_{a1} = \frac{E_a}{jX_1 + jX_2 + jX_0}$$

$$\therefore |I_{a1}| = \frac{E_a}{X_1 + X_2 + X_0}$$

$$X_1 + X_2 + X_0 = \frac{E_a}{|I_{a1}|}$$

$$X_0 = \frac{E_a}{|I_{a1}|} - X_1 - X_2 = \frac{1}{1.0033} - 0.3 - 0.48 = 0.22 \text{ p.u.}$$

RESULT

Positive sequence reactance of generator = 0.3 p.u.

Negative sequence reactance of generator = 0.48 p.u.

Zero sequence reactance of generator = 0.22 p.u.